ActInf Livestream #041.1, Extended active inference: Constructing predicti

Livestream | *1:49:54*

All right.

Hello, everyone. Thanks for joining ACT-INFLab livestream number 41.1. It's April 6, 2022.

Welcome to the ACT-INFLab. We're a participatory online lab that is communicating, learning, and practicing applied active inference.

This is a recorded and an archived livestream, so please provide us with feedback so we can improve on our work.

All backgrounds and perspectives are welcome here, and we'll be following good video etiquette for livestream.

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All right. Today in stream number 41.1, we are continuing to learn and discuss the paper Extended Active Inference, Constructing Predictive Cognition Beyond Skulls by Constance, Clark, Kirchhoff, and Friston from 2020.

And today in the dot one, we're going to have some introductions, revisit the roadmap, and some of the big questions of the paper.

And then we can have an open discussion around the figures, formalism, anything else that people think is relevant.

And for those who are watching live, feel free to add any questions or comments in the live chat.

So we can start just by saying hello, hearing perhaps Blue's first thoughts on the paper, and from Dean and I, we can revisit what we've been thinking about potentially since the dot zero, which I'll save for soon.

So I'm Daniel. I'm a researcher in California, and I'll pass it to Dean.

Thanks, Daniel. I'm Dean. I'm up here in Calgary. I'm a little nasally today because I'm fighting off a bit of a cold.

What I've been thinking about since we did the dot zero is the sort of the idea around how cognition is maybe cultural.

And I'll just leave it at that because maybe we can have a bit of a conversation about that today.

I'll pass it to Blue.

Hi, I'm Blue. I'm an independent research consultant in New Mexico.

And I yeah, okay, so I'll give some first thoughts on the paper.

One thing that really stuck out to me was the idea of glue and truth.

So that like gave a whole new like level like so so in order for you to have extended cognition, you have to believe in it and you have to you have to be glued to it.

Like you have to have access to it whenever you need it, and you have to believe that it's valid. Right.

So that that's the glue and truth idea that they unpacked in the paper and it gave like a whole new meaning of like being glued to your phone to me like I was like, oh, this is why we're always glued to our phone.

There's glue and there's truth. Right.

So that was kind of interesting and then definitely like to touch on what Dean was saying, I think that the cognition and bedding and culture is a super interesting aspect.

And like in the very last section of the paper, they were talking about like the ability to give like theater like plays without being able to rehearse just based on cultural cues and knowing the context and just being so well versed in how to cue off of each other or this kind of thing.

And it reminded me like maybe like jazz improv, like we know like what's acceptable or like when you're playing when you have like a jam band and you're playing all the time, like you kind of just know where to pick off or or, you know, Daniel and I are constantly leaving stigmatic traces like like there's like this extended cognition that I see happen, like we'll both be doing the same thing at the same time.

Like I was just doing that, like it's hard to eliminate redundancy in those kinds of circumstances.

So, so there is like the shared culture and how like central, like even because culture, yes, we have culture as a human society, but like within the Elizabethan theater companies, there's their own culture, right?

Of, I don't know how many people were in the theater company, 20.

So, so like there's that and being able to cue off of each other and like, you know, the active inference lab, which is growing every day, reach out if you haven't yet signed up to be a participant.

So like within the active inference lab, like where we have our community and we have like, you know, some people we interact with on a very regular basis and then others less so.

And so we're kind of able to cue off of each other in that kind of way.

And so how does this go out to like your religious culture, your state culture, like your country culture, like let alone society at large.

So, so this idea of cultural cognition, I think is very like the valiance and also like really speaks to the IQ tests or not, not IQ tests, but even like childhood development and learning.

Like in some cultures, when you give a child, you know, some pieces and do a demonstration for them of how to like, you know, stack blocks together or how to like make a Lego thing.

Like in some cultures, like, I mean, especially in the West, you see like a child who assembles the blocks, like maybe starts off how the demonstration happened, but then like does some new innovative stuff.

Like we see a child like that, oh, that's so like creative and what an ingenious way to use these, you know, tools in a novel way.

But in some cultures, like that child must be an idiot.

Why isn't he just replicating like what the adults showed him how to do?

And that's a legit like cultural phenomenon that speaks to intelligence.

And yeah.

Okay.

I rambled enough.

You've got my thoughts.

Awesome.

Well, at least we know there'll be a lot to jump into.

Let's return to the big question.

So for more background, of course, people can read the paper or check out the dot zero.

But let's just pick up with the big question and then the roadmap and then we'll kind of jump into the more unstructured slides.

As we framed it here, the question of this paper about extended cognition is about how cognition is.

And the way that the authors approach that is this perspective of the cognitive niche, which they develop from the perspectives that are focused on here in this paper with this dialectic of the cognitive niche consisting of a psychological habit from ecological psychology.

One can think of as well as a functional habit, like a neuro behavioral habit.

And they also have a selective ecological evolutionary perspective on the cognitive niche that Axel Constant and others have developed in other work.

But in this paper, they were focused mostly on the psychological and the functional and the timescale of behavioral ecology rather than evolutionary ecology.

But this is all kind of in the background.

So they focus on the psychological and functional dimensions of the cognitive niche and use that to sort of navigate and triangulate cognitive extensions.

So any other ways to think about it or just what did other people think of were just like a big question that would maybe mention active inference or not.

But what is the kind of question that leads somebody to want to read this paper and understand what they're conveying?

Yeah.

I think one of the interesting things here is that if you encounter something that you don't believe is an extension or a continuation of the prior culture.

So you're walking into a new habitat.

We'll just call it that.

What does that then imply in terms of your ability to cognitive as a newbie cognitively extend into that situation?

I think that's what this paper is really interesting in terms of discussing because not so long ago, there were a couple of researchers writers, Lave and Wagner, who talked about experiential learning and how you could be on the periphery of a culture.

And you would slowly move your way closer and closer to whatever the core of that culture is.

This doesn't speak to the cultural aspects of cognitive extension in that way.

So my question is, when you walk into a new culture or something that you don't feel that you're necessarily familiar with, how does your cognitive extension get taken up in that new culture?

Yes.

As opposed to the participants in the culture, right?

There's two components to this.

It's how the people will treat you, for example, but it's how the culture treats you as well.

So there's two parts to that.

And I just, I think that's really interesting.

They haven't broken it out in this paper per se, but they certainly lead to that kind of question because of the research and the argument that they've been able to put forward.

Nice.

Three researchers walk into a new habit.

Is it a psychological habit?

Is it a functional habit?

And then that makes me think about like the culture of a music festival or something.

When you're at the periphery, there's people who are just joining like you are, or people who are leaving or people who want a little bit of a space away from the main action, but using that sort of inner and outer culture idea.

Very interesting.

Yeah, Blu.

So the idea of being like a newcomer is pretty interesting in, in this context, because there has to be some kind of reciprocity.

Um, that like you are expecting the niche to be a certain way, or maybe you learn the ways of the niche.

You don't really have an expectation or, or, you know, like how your old niche was and you carry that forward with some prediction.

But as the niche is updating you, you are also updating the niche.

And so as the culture is updating you, singular you, you are also updating the culture and it goes around and around like that.

And I think, um, or, or I wonder what degree we all update our culture.

Like what is the single characteristic?

Maybe it's like a big five personality trait, or maybe it's a, uh, you know, some kind of inner culture.

Some kind of Enneagram type, or maybe there's a single dominating thing that makes you have a strong influence on the culture around you.

Or is it just like the rate at which you transmit information or the number of people that you transmit information to?

I mean, we all know that like those nodes that are hotspots in like, you know, social networks are, you know, like they're the influential nodes that have the billions of connections.

Um, so is it just the number of connections that you have, the level of information that you transmit or the quantity of information that you transmit?

Or, or is there something like some personality trait that actually gives you the ability to update your culture more influentially than others?

Yes.

Yes, Dean.

Yes, Dean.

One last, one last quick, one last quick point.

I want to kind of build on Blue's point there.

So part of this, and I, and I'm, I'm using terminology that's not part of the paper, but I think we could look at, in cult, enculturation.

And I think that's what I'm using as a kind of, a form of domestic domestic, domestic, you domesticate.

Right.

And so I'm not sure when a person walks into a new culture, whether they're perceived as sort of non domesticated, and then we're going to get them caught up or brought up to speed in terms of what the culture is.

And I agree with you, Blue, it is a circular thing.

I just wonder whether there's a certain amount of wildness that's brought in when somebody new is sort of thrown into the, the bigger cultural pot, and whether there's an effort to try to bring them to be more influential on the, on the one, than the one being influential on the many.

So I always, I always, I'm always curious about those things.

And I don't think that this is going to get answered right now, but I think that's an important point to sort of remember, especially when we get into the flipping part here, which I can see Dan has already got the, the graph up.

So let's go there.

We will come there.

It just made me think about when the niche is not other people.

So going on a walk through the forest and the niche is going to be like other trees or just the path itself.

So this kind of figure two representation where there's an active process by which not just as the entity doing action selection with respect to the niche, but the niche is basically being agentified.

And it is taking some sort of active process by which it is doing an inference on the entity.

And in the forest, it's like, Whoa, are the trees alive?

You know, is this Lord of the rings?

But if this is another group of people, and this is your social niche and your psychological and functional habitat is other people, then it is like quite literal.

This could be two people in conversation, or this could be one versus multiple.

And so in the social and cognitive and cultural niche, it is like less of a leap to talk about how there's cognitive processes on both sides.

But then that helps us recognize that actually it is a continuum.

And it's like, okay, so if we can very clearly see how in the conversation, there's this optimization happening on both sides, then what does that look like when we're in the forest?

Okay, cool.

So some of those big questions have been raised, and then let's just revise the roadmap and see where they went on their little forest journey.

And then we'll jump back into the slides and go from there.

So they introduced the cognitive niche, active inference and extended mind concepts, and then discussed the argument that they're going to lead us down.

And the way in which active inference plays into it.

As we mentioned just recently, they're focusing on the functional and psychological aspects of the cognitive niche, leaving for later that eco evolutionary perspective.

And they're using active inference as an integrative framework and set of formalisms to help address that.

They go through cognitive, psychological and functional niches, provide a case study with earthwork,

and have several figures that we spent a lot of time in the dot zero looking at where in figure one, there was somewhat of an active entity in a niche that could have its own dynamics.

But then figure two introduced the idea that the niche is also doing something like what the entity is doing, but with a symmetry.

So same, same, but different.

And then after that discussion, they take the full turn towards extended active inference, EAI.

And they recognize, well, if the niche or whatever is on the other side of the entity in the way that we're modeling it is doing something cognitive,

is that a way in which we can recognize boundaries and interfaces as well as distributed or extended cognitive processes?

And they approach that from a few different angles, like extension through space, spatial extension, as well as extension through time, temporal extension, related to diachronic cognition through time.

And then they also discussed a few pieces related to, for example, epistemic actions and the relevance of how distributed cognitive systems use epistemic actions.

For example, knowing where to look to find some piece of information in the niche.

That's kind of how they went through this paper.

Any comments on the roadmap or we can now turn to the blank slides.

Don't forget about the spiced food because that was also part of the case study.

What about it?

I mean, that was like there was the earthworm case, but then there was also like the case of spicing food as like a cultural practice and how like potentially, I mean, even the act of cooking food or how we prepare food overall, like some parts of food are poisonous or like, I mean, you're going to eat blowfish.

Like that's a very precise, very mentor, mentee transmitted practice.

So, but, but I mean, more than that, just also like the spices that we use can sometimes be like, you know, poisonous agents and they're slightly toxic to us, but not in small amounts.

And, but they also like help to get rid of any parasites that might be residing in the meat that we're eating.

But yeah, they, they use that as a, as a case study also, not just the earthworm with the kidney.

This is a book that we've been reading at home recently.

Much depends on dinner.

It's kind of in the culinary writing genre, not something we talk about too much directly here, but it's a very interesting approach that really does some deep dives on what we could see from our act in the lab vantage point of like extended and encultured cognitive processes.

And thinking about how food gets to the table in all of the ways that it gets to the table.

How do people get to the table?

How do we show up and how does that differ through time?

So it's kind of a cool book.

Similarly, like the omnivores dilemma, which I just read was also like, how does your food get to the table?

Where does it come from?

And yeah, that's a whole, there's a whole set of practices and like, not only that, but it's actually, I mean, you go back to thinking about serval and thinking like a state, thinking about thinking like a state.

There's a whole set of practices that actually construct our society based on the way that we get food, which is really pretty remarkable.

It's a distributed metabolism with bringing resources and nutrients to and from.

And so it is very related.

Okay.

So here's a few of the options.

People can take a quick little vote or state their preference.

We can talk about walking into new cognitive niches.

Talk a little bit about glue and truth in the context of extended cognitive processes.

And culture about applying extended active inference.

And how that is modeled in its deployment, as well as maybe from a experiential perspective.

What does it feel like to be applying active inference in the extended way?

We have the question from Dave about thinking about partitions.

And there's a few other ways we could go, but any preferences from these?

So when we talk about applying active inference, are we talking about this from a realist or instrumentalist perspective?

I just need to know.

It determines my vote.

I think you've already said enough.

What would it mean?

Why does it matter about realism and instrumentalism if we're talking about applying extended active inference?

Like what are those as you're seeing it now?

And what would the realism flavor or instrumentalist flavor look like?

Well, are we talking about actually what is it like to apply active inference in an extended way?

Like what is that actually?

Like it takes on a very realist flavor when we're thinking about extending active inference beyond our own skull.

Or are we modeling?

What are we actually doing here?

Are we modeling collective cognition, collective predictive processing as extended active inference?

Or what are we actually doing here?

I would like to know what you guys think.

Yes, Dean.

My curiosity around this is always to do with things that actually happened, but I still think there's a sort of a, an instrument, a tool that's going to be a tool.

Or instrumentalist part around it too.

When I first joined this community, I don't know that I could have gotten, become part of whatever the community is evolving into without having both a kind of a realism perspective and an instrumentalism perspective.

So I'm kind of, I'm always kind of curious when people want to break it out into those two things.

I'm a minimum of two thing guy.

I think both of them, both of them exist.

And I don't think we can actually do anything without having both of them.

So let's say we drop one of the other blue.

What happens in terms of our ability to sort of get a real grip on or optimize or do any of the things that we want to do in terms of becoming enculturated?

Like if we drop one of the two and just put all of our, our, our resources in the other domain, does like, do we do that naturally or do we do that in, in those domesticated situations?

Right?

If we're doing it in the wild, if we're truly having to go out there and forage, don't we use both?

That's my question.

So I will respond to that.

Um, I do think we use both.

I think I drop the instrumentalist perspective all the time.

Like I drop it when I think about extending my cognition into my niche environment.

And, and I think about extending my cognition into my group of active inference people or, or, or whatever.

I dropped the instrumentalist perspective frequently in language, in thought.

Like I think about, Oh, what is it like to actively infer X, Y, Z, but I never lose sight of the instrumentalist perspective as the real way things are.

Right.

Does that make sense?

Like, so the instrumentalist perspective is the real way things are like, this is a tool.

I talk about it as if it's not, I think about it as if it's not, but essentially in my generative model, like from the realist perspective, it's not.

From the realist perspective in my model.

When I talk about my model, I am modeling my model and I'm doing it with the instrument of active inference.

Yeah.

Well, again, that's, that's where I find this really interesting.

If we're going to extend, extend, it's hard enough to, to talk about building a niche or constructing a niche without tools or without realism.

But if you're going to do that as an extended cognition exercise, now it's even more complicated, even more complex.

How do you drop one of the two and, and, and continue to feel like you're, you're accomplishing something.

I don't, I don't want to get into debate where I don't think we're getting into debates about it's one or the other.

But I think when you're trying to figure things out to just say, okay, I'm going to, I'm automatically going to blind one eye.

Okay.

Just because I want the other eye to grow bigger.

The other eye doesn't grow bigger.

So why are you doing that to yourself?

That's what I always ask.

So we can think about applying extended active inference.

And as blue initially asked, are we approaching this in what ways from a realism and in what ways from an instrumentalism perspective?

So the realist question, just we're at the DMV, we're at the, getting our driver's license renewed.

And they ask us to cover one of our eyes because they actually do want to understand that one of the eyes has a certain visual capacity.

So we're closing the instrumentalism eye and it's like, to the extent that we can even talk about the real cognitive systems actually really do have functional dependencies that extend beyond any actual boundary in the world or that we state.

So however many philosophical caveats you need to add about realism, this eye has no problem seeing that the functional components of cognition are distributed.

So that is realism or really extended active inference.

So R E A I, the instrumentalism eye is how are we going to utilize active in a way that is useful.

And so here with this eye, we can see that no matter what, we're going to have to model along with compromises like accuracy minus complexity, the ways in which our models of the cognitive systems, they actually do require in the model, the as if influence of influences beyond their neural system.

So whether those influences beyond the actual boundaries are real or whether those influences beyond the statistical boundaries are statistically real, we can oscillate and like look at something because we're the one who's generating those visuals and integrating those perspectives into like a single coherent generative model.

So maybe there's even other perspectives, but that won't mean that.

Won't mean that we are.

Not having a singular underlying coherent generative experience.

So, okay.

That bridge being crossed or burnt, depending on what you think.

What are some situations where we would be applying extended active inference?

So, and also to distinguish that question from just like, where would we want to be extending active inference?

Where would specifically extended perspectives on active inference be really important as opposed to just non extended?

And maybe it'd be helpful there to look at figure one and two, just to remind ourselves of those differences.

So think about the examples while we, yes, blue first go for it.

So, in the spirit of active inference being a scale free framework, like I think about extended active inference as very critical to this scale friendly slash scale free way that we think about active inference just because as a biological system, we're layers and layers and layers and layers and layers deep.

So like I, this like whatever my ego perceives as me, blue, I am an embodied image of extended active inference.

So, so, so I don't see it happening anywhere.

I don't see it extended active inference, not happening anywhere.

Like I see what happens in my skull, a function as of what's happening in my cells and my tissues and my organs, et cetera.

Because all of those sub units of blue are doing this extended active inference with one another.

But if, if we want to look specifically at like a group function, which, which may be necessary or, or maybe not, I mean, to go beyond our skull, are we looking at like how organizations function, which maybe Dean wants to talk about more or how, you know, states function or how institutions function at large.

So, so, so do we have to get actually beyond the physical body to go beyond our skull or can we go within our physical body to talk about extended active inference beyond skulls?

Yes.

So, so, um, we can think about like that great point, which is that nested systems of which we often talk about can actually be thought of as extended active inference systems.

So, so we've talked about like ant nest mates in the ant colony.

And so there's the sense in which the lateral interactions from one nest mate's point of view, because those lateral interactions are with other nest mates, there's cognition on both sides of the interaction.

So it is like an extended active inference process.

And then also there's the extended active inference with the physical substrate and the stigmar G and the building of the nest architecture and the pheromone deposition.

And that kind of physical substrate is more on the mere active inference rather than adaptive active inference end of the spectrum.

And the physical substrate reminds us more of figure one where you have an active cognitive entity that's doing some sort of free energy minimizing process that's helping it select policy.

That's the argument.

And then it's feeding into the dynamics of the external niche, which can be complex and they have their own endogenous change dynamics, which is seen by the fact that like the rate of change on the external states is a function of the external state.

So the current state is a function of something involving the current state of those world states plus a noise term.

So this can be a complex recursive nonlinear function, but that could still be a mere active entity because it's not actually engaged in a policy selection, rather just a complex dynamical unfolding.

And then in contrast, in figure two, we see that the external states are still taking similar arguments, eta and a.

So the current states at that time and actions feeding into it.

That's what's influencing external states.

And we see those arguments are still here.

Actions feeding in from the entity and the current state of the external states.

Except what it's doing is no longer just the rate of change is a function of plus noise, but an active policy selection more like the kind that the entity is involved in.

So maybe figure two is reminding us more of like two nest mates interacting with each other and figure one is more like a nest mate interacting with the physical stigmatic niche.

And so in that way, it opens up all of the discussions we've had about multi scale active and nested systems and thinking about them as extended active inference.

Dean?

Yeah, in the paper.

I'll just read one sentence from the paper.

It says here, the concept of the cognitive niche that we refer to here is a sort of hybrid between the concepts of the selective developmental and cognitive niches.

It's right near the beginning.

Yeah.

Yep.

Right there.

Yeah, you got it.

So in the context of what were those two figures there.

When we're talking hybrid, what are we really talking about?

We're talking about the comparison of say the electric component of the engine and the internal combustion engine that are both in one car.

Would that be a good analogy in terms of being able to figure out what part of this is the vessel, what part of this is the electric motor and what part of this is the internal combustion motor?

Would that be a good metaphor for what we're talking about?

What's the car?

Right.

Well, that's, that's the, is that the culture piece?

I'm not sure.

It reminds me of that.

Yeah, I don't know what a cognitive niche is here.

Like, I don't know if I want to entail culture with a cognitive niche, because this is what this paper is about, or whether I want to entail cognitive niche with culture, which is something that we've talked about in a whole bunch of papers, right?

So that's a good question.

I don't know.

It reminds me of your earlier statements about how, like if a pizza is something with toppings on top, and then if you put it upside down, is it still a pizza?

Right.

So is car the total functional phenomena involving vehicular transport? Or at what point does that also require like gravity and a road and all these other pieces that are part of the extended process by which cars come to be?

Like, is the factory that made the car part of the car?

I mean, it would be interesting if they just blinked into existence, but because things are part of process ontologies, how can we connect this question of extended function to the process and object ontology angle?

Okay, cool.

Okay, cool.

Blue, I see some stigmagy happening.

Okay.

So, to kind of close this little thread on applying extended active inference, though, of course, we welcome a thousand more specific contributions and we hope to continue building these actual models and examples and interactive dashboards and

everything out.

There are multiple ways to think about what is extended and probably even for consensus on the same perspective on the world as it really is, or even on the same choice of models, there would still be a lot of degrees of freedom in how we actually apply extended active inference.

But one tantalizing cue there from blue was connecting it to the scale-free, scale-friendly components and potentially connecting what we've been talking about with active inference in nested multi-scale systems to this perspective on thinking about their niche as consisting of other mere figure one or adaptive figure two active entities.

What do we see with the termites?

So, I just was thinking about like the literal pushing off into the niche in kind of like this scale-free slash scale-friendly way.

I saw the last author of these papers give a talk at the mathematics of collective intelligence workshop and it was so cool, like really literally like the termite mound.

And I'm sure that the ants' nests do the same thing, Daniel, but I think that they're not very studyable because as soon as you go in to start to try to figure them out, like they fall apart.

They fall apart. Is that, that's true? So we kind of get a better...

Well, many termites build underground and in logs, so it can be hard to assess. Above-ground termite architecture is of course amazing.

And more ant architecture is happening underground, which makes it really hard to study unless you have like a high-resolution landmine sweeping device.

Landmine sweeping device. So like, are there like images? I mean, yeah, I don't see how to do it really without destroying it because they're like made of sand, right?

So how do you like... I mean, I remember like when we were kids, there was like the frame, the two glass panes, and you would get some ants and like throw them in there and watch them like dig tunnels.

But like other than this, like very artificial environment, like the deer on the trampoline, right? Other than this very artificial environment, you know, in what way do we know about the ant architecture?

But, but my point was actually just that, that the termites really like, you can predict the behavior of the termites based on like the velocity of the wind and how hot or cool or like they build all these different structures.

And you can predict like, and they not only build these different structures, but they will actually like deconstruct their mound and then reconstruct it like based on whatever is shifting in the environment.

And so this latest paper is cool. Like really shows the, it gets into like the math behind how termites, termite mounds are constructed, like under what, you know, wind velocity and so forth.

But, um, I just thought that that was a really neat, uh, example of literally putting the offloading into the, I mean, we literally do that as humans.

And so we're pretty familiar with like, you know, the smartphone and the different ways in which we do cognitive uploading, but this idea of cognitive uploading, which I've always used cognitive offloading, um, cognitive uploading into the niche can kind of tell others, even future others.

Uh, what was happening to you in your life.

It's like, you're writing your own little, you know, history book as we are cognitively uploading these videos onto YouTube.

You know, this will be here after we're gone.

Uh, but the termites do this too, in a very cool, like physical way.

I just thought it was neat.

Tucker has corrected me in chat.

It's not a landmine sweeping device.

It's an ant mind sweeping device.

So thank you, Tucker.

Tucker.

Cool.

Yes.

I think, um, extended active inference systems to study could include starting with a colony like systems or various other systems.

So what would be another good place to now jump in as we sort of walk back to the midden pile, walk back to the trash heap of ideas and then figure out where to repurpose and go back in.

Um, maybe you meant, yes, Dean.

I just have a question for both of you.

Cause so when I think of offloading and they give the example, and I think I put the image up there of the brain on top of the, of the phone.

Um, if we, if we upload and we're talking about intergenerational stuff, are we uploading into culture?

And is that the difference?

Is that the primary difference between offloading something so I don't have to remember it and uploading something so that rules are introduced that you will sort of be swept up by if you join this culture?

I'm asking cause I'm not really sure.

I, I was trying to figure that out myself.

What's the, is there an arbitrariness to this or is it really kind of clear what the line is in terms of they tried to explain it?

Is it there?

They said it's akin to cognitive uploading, but it's different.

So where did you guys come away on that, on that question of what, what's different about it?

Yes.

Great question.

Let's look at what they wrote and then let's see where we can take it.

So they wrote cognitive uploading is akin to cognitive offloading in the original theory of the extended mind.

However, in contrast to the traditional notion of offloading, the notion of uploading refers to the creation of novel cognitive functions that are taken on board by the cognitive niche per se, instead of being merely managed by the cognitive niche.

The functions offloaded when individual agents restructure their world.

So as to minimize internal processing costs and or increase reliability.

So offloading, like you could offload your heavy, heavy backpack and just drop it on the floor.

So offloading has more of a sense.

Um, at least hopefully this is a line with as they're describing it of like, it is allowing the shelves in the world to your offloading what you're carrying into places in the world.

And so the cognitive onus and the imperative for epistemic action is still resting in the entity of interest.

So cognitive offloading could be consistent with a worldview.

That's like solipsistic.

Like there's just this one cognitive entity and they're like offloading, they're off gassing.

It's their exhaust thermal waste.

They're offloading cognition to the niche.

Cognitive uploading takes on more of a sense that there's actually something active on the other side, because like if you're going to upload a file to a cloud folder, there has to be another server that's actually recognizing your packets as you're uploading that file.

So uploading is more consistent with the framing that they have, for example, in figure two, surprise, surprise.

Whereas the cognitive offloading, um, could be consistent with figure one.

Like I'm trying to have this cognitive function of this symphony of bells being played.

And so I'm going to set up these dominoes and these delays in the niche and where to put the sheet music and where to put the post-it notes offloading the cognitive offloading.

So, you know, they're not just uploading the cognitive requirements, making it extended, making it distributed.

In figure two, it goes even a step further.

And that's what they're trying to highlight the distinction with, with their usage of the word uploading, because there's actually something that's cognitive that's accepting the download.

Or the upload from the entity is the download from the server, so to speak.

So sorry if that's repetitive, but I think it's a subtle and important point.

Dean?

Yeah, I think it's a super important point too.

And then, then for me, the question is, so let's say that I have an academic culture that I could upload to.

And I also have a professional culture that I could upload to.

And let's say they're two different things.

Right?

So I've got, I'm a part-time student, and I'm also working full-time at my job every day.

Is there some way of being able to find myself or find examples of myself not being downloaded on per se by each one of those different cultures, but me uploading into both of those and retaining my sort of hybridized sense of the world?

So from a psychological and a functional niche perspective, what does that entail for me as the person who's now kind of doing this oscillation back and forth between these two niche constructions?

Right?

Now they, again, they don't go to that length here, but I think, I think it's important based on what you just said, Daniel, for us to remember, it isn't just a single culture examination here.

The whole point is, is that there are multiple niche constructions happening all the time.

And it's just a matter of whether or not we recognize the differences.

Right?

So I don't know, again, I'm not sure that they answered that here, but I think we're getting closer to being able to find that answer because of what they're, what they're bringing to our attention.

Lou, go for it.

So.

I'll spin up some slides for you, Dean.

So just on that, on that point.

When I really was thinking about cognitive uploading, it's like, it's like putting something into the niche for others to use.

And reuse and maybe interact with, or like remix.

And what you're saying about like this hybridization between academic and professional really makes me think about complex systems and like the Santa Fe Institute where Daniel and I met and the kind of siloed academic experience that maybe people are still having, unfortunately, but that, that people have had.

So if you are in, you know, mechanical engineering versus biology versus economics, like you have very different knowledge acquisition and like a very different culture and a very different way of interacting with your peers.

And there's this kind of like siloed culture, even within an academic culture.

So like even subsections of academia tend to be siloed.

And when you bring people into a complex systems kind of environment, there's this amazing like ability for new ideas to generate.

Right. It's like this cross fertilization or cross pollination.

People call it different things, but, but when people come together with academic expertise, some, you know, shared ability to read, write and do some mathematics.

So when people come together with and intermix their own siloed knowledge, which is deep, like it tends to be very, very deep.

Like if you've got, you know, PhD or bachelor's degree or master's degree in, in your field, like, you know, a whole lot about that field and probably not a lot like about the other field.

So when you come together in a complex systems kind of way, it enables people with this deep, deep knowledge and understanding of their own thing to then interact and it's exciting and it builds new things.

Like I think exponentially, like, like, you know, if I get together with a bunch of other data scientists or a bunch of other bio people or neuro people, I don't have that same kind of excited, brainstormy, jittery idea.

Gush.

Like that was so professional.

But, but generally I think that that happens in this cross fertilization or cross pollination way.

And when I think about cognitive uploading, I think about like you push it in so that other people can interact with it so that other people, so that it can do the cross fertilization thing.

And maybe that's just kind of my naive interpretation of what they were saying in the paper, but that's kind of what I got.

And then Daniel made us slides.

So perfect.

I talked enough.

I totally agree.

One slide's worth in Kairos time.

Totally agreed though.

Like uploading as opposed to, or as distinguished from offloading offloading, you just need to trust that the shelf isn't going to break and someone's not going to steal your backpack while you're gone.

But cognitive uploading, like you got to trust that file is being stored and that that is going to be able to be executed potentially even outside of your awareness or control.

So when we're in like a interdisciplinary context or just sharing like cognition in a team, we're getting it done as a team.

So the uploading brings in this whole element of like trust and culture and glue that others are going to be able to do things that you don't want to do that you can't do that you wouldn't see of doing all these different ways in which their active cognitive model is different.

And importantly aligned and importantly, contrasting and complementary it's all of those things.

So here's a few visualizations of borders between or interfaces between entities and each other or entities in the niche.

So on the left side, it's like the overlap between two systems, whether we think about them as like, oh, that block in the city, it's part of little Italy and downtown.

So there could be like a block, there's a border region, but it is both included in the two sets, or it can be a little bit more blurry or fuzzy around what those orange and blue sets are.

But still, there's like a region that is overlapping and claimed by both.

There's another case where the border or the interface is something that is like overlapping to both but also a distinct region, perhaps like a cell membrane, or in the full separation case, it becomes its own separate region.

And that entails now this recursive boundary setting because now there's Wall Street, there's the street in front of the wall.

And so this just made me wonder when we were talking about academic and professional, what kind of synthesis or what kind of integration are we looking for?

Which one of these setups is predisposed towards effective extended active inference?

Or is this something that we determine about how it really is in the world?

Hashtag realism.

Is this a model selection choice from an instrumentalist point of view?

Can I just speak to that for a sec, Daniel?

I think this is interesting because in this example, and also in the paper where they talk about desire lines, we do have a tendency to sort of see these things be at the border or the artifact of the number of humans or animals that walked to the watering hole, and the result on that, and they speak to that clearly in the paper that

So sometimes that is not acting in the environment is not acting in a surprising way.

That's the result of what ends up happening when a whole bunch of people follow the same path or use the same stones to break the nuts.

That's the capuchin monkey example.

Or use the same spices, right?

In a particular thing.

One of the things that we kind of just touched on briefly in this dot zero was, so what if you took all of those examples that you had of boundary, and then you can see that.

And you used green as a, as one of the things to sort of show it in the rectangular shape.

But what if you were able to twist that and do with it what we have when we, for example, use the, the Mobius strip in the keywords.

Do you have a representation of that here?

And why would that matter?

Because I think that actually really does matter.

And I think it's something that we tend to overlook.

So if we're talking about, I mean, if we're talking about transdisciplinary and blues point, I think this fits really, really well.

But if you wanted to talk about people walking into a situation who are creating a new cognitive niche, they don't come from a place of knowing.

They basically come from a place of not knowing.

And they're going to build that new boundary or that new border.

And they're not going to be restricted to the oval shape.

They might be able to put a single twist on it and come up with, like, like I said, the example that we have that we put up as a representation behind our keywords, or maybe they'll do a Mandel-Brodian two cranks of the system, and they'll come up with the two loops that are orthogonal to one another.

Right? And what does that mean in terms of a representation around uploading as opposed to offloading?

And I think that that's, yeah, exactly, as you're building it out here.

This is where it actually gets interesting because I think that's where, again, if we always see ourselves as where does my cognition end and begin, we will get one, out will kick a certain number of representations, all conforming to that pre-

We desire to keep the cognition contained and framed and stabilized.

Now, if we look at this from the environment position and the argmin flip or inversion of that, can we just limit the inversions to the two sides on either side of the Markov blanket?

Or could we actually look at the Markov blanket twisting and reforming itself?

Because, I mean, it opens up the space for that possibility. Now, whether we choose to give any attention over to that or make ourselves available to that, it gets really, really interesting in a hurry.

So, I think it is really interesting. And where we put the boundary maybe depends on what we feel that we are uploading or offloading.

Like by us uploading these video series into the niche environment of the internet, we, other people are able to cognitively offload.

Like they're offloading their need to delve into deep read and unpack this paper.

Like they could just, you know, they've offloaded it to us and they're just watching the YouTube video.

Not everybody. Sorry if you, if you're really actually reading the paper and really into it and just want to, want to hear someone else reflect.

But in many ways, you know, we provide a summary that, that in today's information, late in age, it's really impossible to just read all the papers, even in your own field when there's five papers released a day.

So I don't blame you for doing some cognitive offloading there, but by us uploading that enables someone else to offload.

Is that always the case? Is there like this interplay with upload and offload?

Like I think about Google maps, like that is the biggest cognitive offload that I've seen in technologically, in technological advancements in my lifetime.

Like I no longer remember how to get anywhere.

Like maybe if I drive there 15 times, maybe then, but, or maybe if I know someplace next to it, like, like, but if, as far as driving to a new place, a place I've never been a new housing development, for example, like I have a friend that moves in and I've never been there before.

I'm at their house 15 times before I remember how to get back and forth.

And I Google that myself the whole way every time.

Uh, it's massive cognitive offloading, but only because Google was so kind to upload the information only due to uploading.

Can I then offload?

And then does that also work culturally?

Like in terms of, I think about.

Institutional or hierarchical organizations, or even thinking like a state, I have someone who will make shoes for me.

I don't need to know how to make shoes.

I just need to know how to access shoes.

I don't know how to make shoes.

So, but because someone else is uploading that information technology, their product to our culture cognitively, but it's like the output of their cognition.

Actually, I get it.

Maybe they could actually teach someone else how to make shoes if they had to, but, but my point is, is, is that there's like this interchange does upload somewhat because you upload it to someone else download it and then they can offload because of their ability to download.

Sorry if that was rambly.

No, I think that's, I think that's perfect.

And my question then is from a learning standpoint, are we more adept at accepting other people's information because that expedites, we can follow and, and, and use that.

Your first example, right?

Well, here's somebody Google, I mean, for Google, it's great for them.

They can update, they went and figured out how to become the go-to source for uploading.

But what are we doing in terms of giving people a sense of their agency over uploading, as opposed to simply just following and being consumed by the Googles?

If we're going to invert this, we need to ask that question around what learning, what we're giving people in terms of agency and, and knowing the difference between the trade offs.

And so that happened with, you know, you're not building any desire lines.

You're, you're, your GPS is doing it for you.

Now, now you feel completely dependent upon that thing.

And you don't have to be able to make shoes to be able to understand the difference between what you might need in one set of circumstances on your feet, and what you might need on a nut in another circumstance, set of circumstances on your feet.

But are we drifting to that place that says, I don't even know what shoes to put on today, I better go and find out from Google.

Yes, very interesting.

Keeping with the uploading offloading desire paths and Google.

Thanks YouTube, by the way, for the affordance.

What if each person were only seeing locally which paths were attractive to them, which search results were appealing to them?

But then there was an eye in the sky that could see the statistical distributions of everyone else in their uploading and offloading.

That would actually make that a tremendous resource for the eye in the sky.

And so that's a little bit of the asymmetry, but the challenge and opportunity.

Also, the caption of figure one does explicitly mention like action, perception and learning.

So that's very interesting.

And it does raise the question of what is the difference between perception and learning, which is discussed a ton in active and in the textbook, what the similarities and difference are like with perception being over faster timescales,

but being fundamentally similar to learning, but learning being about cognitive parameters and usually updating over slower timescales.

And then what does learning look like in a cognitive offload world?

In the no offload world, then in the cognitive offload world, and then in the cognitive upload world.

So just to kind of maybe just give one example, the no offload world is the test where you can't bring any notes in.

Yeah.

In that case, it's like what you know about organic chemistry is in your skull.

So one could go a little bit broader than that, but you can't take in your notes.

The cognitive offloading, which is going to beg the question of like epistemic action is you're allowed to bring in your own notes or even the textbook.

So it's a dead artifact in the niche.

It's a mirror active entity.

It's an epistemic resource in AOS.

We would call it an informational entity in the niche, and it can be referenced and you might only have to learn where to look, learn where you put your backpack, learn which chapter to look to when you want to resolve a certain kind of uncertainty.

So it's an affordance you didn't have in the no offload case to have the textbook or the notes.

What's the cognitive upload?

Is it phone a friend?

Is it go in with the chemicals and have a lab kit?

Like, what does it look like to go one step beyond?

Yes, you can take your notes into that organic chemistry test.

What is the upload?

What do you think, Dean?

I think the upload is when you are taking the information that you've received here.

And finding yourself migrating in a completely different culture.

And somebody in that completely different culture says, I want you to solve this problem for me before I can hire you.

Can you do this?

This thing first show me that you can do this thing or resolve this this issue for me.

So it's it's taking information from one context and finding yourself in a completely different context, not from the person who's decided to be the person.

Who's decided this is the amount of material that we've covered in these past 13 weeks.

And now I'm going to select out the points that I want you to, as you said, load into your into your bucket and carry into the testing situation.

I want you to actually now to take what you've learned and I want you to be able to go out back into the wild now where you're going to have to be able to solve something as a result of you loading loading up your your quiver or your your 13 weeks worth of information.

Right.

That's the that's literally finding yourself now working with a new culture and that new culture taking on all the properties that they're talking about in this paper.

Another angle on that.

Oh, yes, please.

So my take on that would be to, like, say you have all of the material that you learned, you have all the books, you have all the notes, you then go teach it to someone else, and they go in and take the test for you.

So that would be uploading.

Awesome.

Awesome.

Is that what you were gonna say?

Yeah, so we have too much.

We do too much.

I was thinking of a driver's exam.

It's the difference between the no notes driver's exam taking the test at the DMV.

Can people tell that I just renewed my license?

And then there's the having the notes, and then there's the actual driving.

And so that is where the functions are no longer hypothetical and in a testing scenario, but they're enacted.

And that means that it's different to say you should do this when you're parallel parking and it is going to be 15 feet versus the actual enacting of it.

And that cognitive function really or instrumentally is happening in this extended way with the visual components of the car and the proprioception of the car.

So that is one angle is towards the practice bent and then blue awesome point with taking it towards the teaching, which is like if you had to send a proxy and what if we all had to train a proxy text test taker?

It would take like, well, you know it when you can teach it to another level because that would entail many, many aspects of making sure that you could upload it in a way where then they could offload or not.

And then of course the infinite regress is what if they had to go train somebody else and then you would have reinvented academia from scratch.

Dean?

Dean?

Dean?

Dean?

Dean?

Dean?

Dean?

Dean?

Dean?

Dean?

Dean?

than the one that's offloading which there's way more offloading in the teaching realm than we

probably care to acknowledge or for the um plot embedding offloading is more like instructionalism

you're offloading the recipe and then interactionism requires the other interact ant

to be a cognitive entity and so that's more like the figure two it's more like cognitive uploading

and the more interactionism is at play the more figure two like we're in a space of the more it is like a question that dave has just asked in the chat which is isn't the result of offloading a new goal-seeking self-maintaining autopoetic thing all caps and and telchi that's kind of aristotelian a little bit i believe um yes and that's a little bit like um our progression here here was two two things and their interface and then it got more and more shared or delineated or less until at some level of interfacing and interactionism and extended cognition from a functional or psychological perspective which is what they explore in this paper or from an ecological and an evolutionary perspective like constant et al explore elsewhere it does become a new kind of thing the eukaryotic cell is a thing and it's a thing that involves these nested and interfacing systems and then i took the colors that we had here and overlaid them from this abstract space into this kind of space that we've seen with the figure one and two representations and isn't that interesting that the blanket states are the interface the border states and so as we discussed the dynamics even in the mere actinth case could be very chaotic or very difficult for the entity to predict so mere actinth shouldn't be seen as meaning that it's like simple or it doesn't have rich dynamics how do we know which of these settings we're in or how do we make the best of these settings wasn't there daniel can you find the slide in the 41 deck where we had we talked about the joint self-evidencing i think i i guess i'm i i guess what i'm saying is is that joint self-evidencing on one level of instructionalism or the joint self-evidencing in terms of the interaction piece um i think that that that's you you can decide how big how how how sophisticated how complicated you want that joint to be but i think that that's what this is essentially saying is can we get ourselves to a place of this closest like the closing of the circle of causality part right that's sort of circularity piece of this when do we when do we realize that it's circular is it when we're at the instructionalist level or when we're at the interactiveness level hello do you have a thought or there's a little bit more slide play that could happen i think to resolve dean's question uh go ahead and give it over to dean i'm just um thinking let's say that we were in a forest of f2s so we're no longer thinking about just nestmates interacting but now we're thinking about like each one of these orange so we're gonna our system of interest is going to be the orange um guys we got f2a f2b and f2c and they're like different evolutionary replicators of a similar kind there are different organisms the blue could be the same physical niche it could be each other in an adaptive niche they could each be in their own battles but their fitness is being relatively compared in terms of their success now what does it mean when on that side that we were just on on figure two when uh they mentioned this closes the circle of causality in which the nation phenotypes are trying to learn about each other to minimize their joint free energy or surprise okay yeah so what does that mean that the causality has been close so one world view would be well let's just look at how the three orange entities are connected and think about how they're in competition and so that might be a totally valid and important way to look at it but um this is kind of like there's various statements i'm sure various phrasings but i won't even try but what is the good in doing so well in the orange f2a free energy minimization such that this whole system is actually maladaptive so that would be like this entity being greedy in this new kind of interaction or thing and so the one uh the orange a b or c that's going to persist the best or have the highest fitness

is going to be involved in the best free energy minimization of this whole system where there's
sense and action coming both ways because just to short term do a free energy minimization here
at the cost of maladaptive actions coming in the future back from the niche is not a sustainable strategy
and so that's like where they tuck the little bit the last thread gets tucked back into the ecology and
evolution and the selective niche which is the one that actually provides at least an adaptive starting
point for the cognitive functional and psychological habitats that they're discussing
what do you think dude well so so again here's the thing if we're talking about just the perception
perception piece and we've got three three versions of perception is that what you're
you're trying to bring out here three similar kinds of entities that are involved in
in fitness but we're focusing on the we're focusing in on the on the perception aspect of that are you not
are we looking at the cultural piece of this when three perceivers
are proximal to one another in in in in an environmental cluster or in a cognitive niche
how do those perceptions align with one another and what's the time frame right what's the diachronic
of this i just want to make sure i'm clear about what you're what you're trying to show here
there's probably multiple ways to think about it or show it let's let's try to make it one one little bit clearer
okay here's going to be the blue niche and now we're going to have those cognitive entities sharing a niche
so so here's going to be a bunch of brains that are going to be sharing the same blue niche
okay so how does this affect what you're discussing they're all being interfaced with a common niche
and so in what way can we say that they're involved in joint action perception loops well that's that's
what i'm that's where i'm where i'm curious because i think it is what this does is it shows that there is
something called an extended an extended aspect of this of this active inference frame but i wouldn't
i don't know how each one of those perceivers realizes what the others are perceiving as well i think
the environment or the niche itself knows collectively what that is but i don't know that each one of those
individual representations of the brain necessarily does know and i think this is where this it trips
up a lot of people because oh i don't believe in extended uh extended cognition because you can't
pause daniel you can't possibly know what dean's thinking and dean can't possibly know what blue's
thinking but eventually as we work out together through communication through some sort of signaling
processes eventually we we kind of land on it on some sort of generally agreed to sense there's a collective
sense around this as well and so do we say well that collective sense isn't cognition
i we don't know what it is but it's not cognition but of course it's all based on cognition
but we're not going to call it cognition because we can't possibly know everything that's going on
in in that other perceiver cognizizers way of looking at blue at this blue rectangle right so sorry
sorry blue night blue rectangle right so i again i this is that's yeah thinking through other minds i
don't know how i don't know how we i don't know how we quantify it but ness but also recognize that it is
is it it does exist and if we get if we get too too um concerned that if we can't measure it it's not a thing
then we'll just pretend like it's not there but it it's there not all things are necessarily measurable
the way that we can hook up the the cap and make sure that it's picking up everything in in the whatever
it is whatever the heck the one is that is the cap as opposed to the fmr i don't know

eeg thank you sir yes um i i just looked up like opposite of extended and so there's like retracted
reduced shrunk narrowed shortened
and it's interesting how extended cognition feels like a reach yet defending the other position
is almost too narrow yeah yeah blue so i just was um going back and reading the author's definition of
uploading versus offloading um and they said in contrast to the traditional notion of offloading
the notion of uploading refers to the creation of novel cognitive functions that are taken on board
by the cognitive niche per se instead of being merely managed by the cognitive niche a function
is offloaded when individual agents restructure the world so as to minimize internal processing costs
and or increase reliability uh for example by posting the yellow sticky note to as a reminder a
function uh for example that a function is uploaded when social and technological change means it is now
taken care of by the niche rather than the individual for example most agents store phone numbers in
smartphones instead of bio memory so the whole number storage function has been assimilated into the niche
uh the niche into which the function has been uploaded can then be passed on to future generations for them to leverage
share and finesse that function and so i wonder if the difference between offloading and uploading is a transgenerational difference
so like i do offloading um like you know with my yellow sticky note or or like my notes in my little pad but i do offloading when i you know
create a publication from those notes and publish it into something that's then archived that the future generations can access
that's that's awesome um here's a thought on that uh connecting it to the epistemic actions so offloading
just like they describe it says um a yellow stick note on the front door to remind them to pick up milk next time they are out
it doesn't say this means pick up milk they've offloaded their cognitive backpack but only they
know their recovery seed only they know the password to speak friend and enter into that extended cognitive
function whereas if it were cognitively uploaded i made an app that reminds people to get their milk
when they're low that is actually something that because of its intrinsic and more integrated actual
cognitive function like the ability to assess information like the level of milk in the fridge
and then take an action like remind the person or not it's a bona fide cognitive entity it's more like
figure two more like uploading than it is like figure one offloading and then that app can be leveraged
and shared and intergenerational in a way where it would be extremely difficult and only through like
implicit learning and seeing it many times would somebody be like ah yes we put the yellow post-it
note when we want to remind ourselves later so interesting example and it's very true what they
say about future generations dean yeah and i think when we get into that that diachronic piece
we're not just we're not just offloading remote remember to get milk what we're doing is we're
when we're uploading we're talking about rules especially if we're talking about culture it's the
old that's not the way we do things around here kind of thing so i think there's there's a there are
differences between offloading and uploading and i do believe that rules because rules are always kind

of flexible they're always a limit around where when we talk about scale friendliness there's an
effective limit around how a rule will be able to function at scale so there's there is going to be
a certain amount of arbitrariness when we get up to that rules place and as we pass rules down from one
generation to the next i'm sure both of you can think of the times when your parents told you this
is the rule and you both looked at them and went what the why is that the rule oh okay i'm so now when
we're talking about rules and we're talking about uploaded things i think there has to be a certain
amount of participation in that rule setting it's not always the case but i think in order for it to be
adopted and actually to be a part of the cognitive niche we're not just talking about content here
we're talking about consequences here as well and so i think that's where this gets really really
interesting in terms of building or constructing that psychological and functional niche but how also
appreciating that it's kind of rules based and it's joint self-evidencing as opposed to well daniel
believes in in jitsi right so everybody's got to follow daniel's rule but that's not kind of how it
works when you bring a bunch of people together and you're trying to come up with some sense of where
it is where it is where are the cognitive where are the cognitive limits here that arbitrariness
aspect of it is actually a part of the game so so that is awesome it connects the intergenerational
niche inheritance to the interactionism and to that idea of having to teach somebody else to take your
organic chemistry test is the point of rule following in the short term from the age of you
know zero to six years old to minimize the perceived number of times that a rule is violated or is it to
minimize the amount of times that your grandchildren violate that rule or for example and so the further
one looks into the future the more interactionist it looks and the more autonomy potentially provided
to the recipient and that might also connect back to mike levin the discussion on training and learning
because the trainer and the trains that may be a functional dyad but it may not be able to result in the
trainee being able to be training somebody else whereas somebody who has learned has a generative model up
date and so they may be able to teach and then also to connect to um uh dean's uh let's uh it was
academic and professional settings right and we talked about like the hybrid models and about um okay so we
had the we had walking into new niches then we had the professional okay here is us and now we're engaging
we can think of this in of course multiple ways so one of this would be diachronic so the diachronic
interpretation would be grandparent parent child this the synchronic but extended spatial interpretation
would be this could all be happening in the same time here's the academic context academic niche
me in the academic niche professional niche other professionals in the niche
so we have through space and through time just like they explored some of those dimensions in the paper
what do either of you think about that the representation i think it's like like this
extended active inference it it grows right it it blossoms so i'm liking where it's going
so it just made me um think about like so i had brought up the temporal um you know maybe to be
uploading versus offloading there's some kind of temporal persistence but also like what if

to be uploading it has to be accessible even in the same time because we don't really know if our youtube video is going to be around in 10 years or watched by anyone else except us or or we don't really know um but but we do uh know that it's watched by like tucker was watching it i guess and dave and and so there are some people watching it so so if uploading just means beyond personal use would that be fair like is that is that uploading um versus offloading i don't know um but then if i
if i remind my daughter like so so it just makes me think of like but what about this but what about this so if i remind my daughter like hey don't forget your lunch or hey you know you need to study for your spelling test if i remind her or if i create a reminder for her is that uploading or offloading
here's your lunch here's a post-it to remind of lunch
how will you remind yourself of your lunch
and it probably there's probably ones that we could add above here is your lunch um here's the food in your mouth here's the airplane flying in and the food in your mouth
or even before here's yeah well before here's your lunch i need to remember to remind my daughter to get her lunch so there's like how am i gonna remember to remind my daughter to get her lunch
like there's that whole like second tier of that yeah yes yes um and this is kind of like a continuum on the um well this is a preparatory uh preamble but then it's like a continuum again from the more instructionalist to the more interactionist um and it probably gets more
allegorical and it gets more flexible it gets more empowering but if you say and you mean it how um
i'm gonna let you decide how to remind yourself of how to get your lunch
but also if you forget your lunch you know it's gonna be really bad then that creates a cognitive tension and so that's kind of like where it's okay to experiment and fail and interaction
you know to improvise one must like make some notes they didn't intend and then continue to roll with it as their new state and just continue on like in improv versus if it was just about playing the sheet music as written then every time a mistake is made one could just push the rails back in
so i think what you're saying is i need to let my daughter go hungry when she forgets her lunch because right now if she forgets her lunch it's really bad for me because i have to drive like an extra hour to go home and get the lunch or it's either way it's an hour out of my day i've got to stop at the grocery store and assemble a lunch it's not just lunch it's like lunch plus like six different snacks that she eats throughout the day so she has to bring all the food with her every day so it's really bad for me if she forgets so um because i end up taking care of it for her i should just let her starve and then i will have uploaded that task
cool yes without getting between mother and daughter can i just say one quick thing here
um i think again if we're if we're just trying to figure out what the extended cognition if there is if there's evidence for extended cognition it's probably in the rule and the consequence so it's to your point blue if there's a consequence here who has to take that consequence on the older person who who who maybe whose prospective memory didn't work out that day and didn't provide all the reminders or the younger person who now is has the effect of not having something to eat but i think the point here in terms of can we the argument because they call it an argument this paper is an argument the

argument for extended cognition would be one where if we upload we're we're probably going to be uploading rules and the consequences of those rules are implied versus the uploading of content and material that tends to overwhelm us and that we use devices then to act as a way of being able to store because our own memorial capacities are are somewhat limited like i mean when i compare my memorial capacity to you and daniel i i can't hold a candle to that right so i should be the one using all of the instruments for that but what i do is i have my own way of sort of compensating for that by developing my own rules sooner and with with a more um consequential nature to that right so that's what one of the things we could possibly be teaching six to eight year olds not just here are the rules but what can you do in terms of helping me who's probably got less capacity as i get older for remembering all the content so there's consequences for me now and there's consequences for you we both suffer a certain consequence in this new environmental niche that we're constructing now maybe the eight year old doesn't give a rat's back say about that she's just hangry now but oh well right i mean so there's going to be some learning one way or another and it could be through guilt or it can be through some kind of a conversation where you actually do introduce the rules and the consequences i don't know in comparison to memorizing the rgb values of an image like the pixel colors none of us have memories that hold a candle to even the smallest jpeg so render onto the jpeg that which is jpegable and then what will that leave humans to do and we hear about cognitive overload cognitive security of course what does that actually mean for designing informational niches cognitive niches from the psychological functional ecological selective perspectives that are not overwhelming or that are accessible or that are resilient or that do stabilize function there's so many awesome questions that this like brings us totally towards daniel i'm sorry but i'm gonna have to bounce out now because uh i've just kind of used up all my energy anyway close this swing i hope you guys can close this thing off for me and i'll look forward to the point too so thanks thank you dean peace all right all right cool blue let's uh think for a few more minutes what could be fun for dot two and just how we can go from some of the rhetoric and perspectives they share towards all of these extremely relevant scenarios that we've been discussing like yeah so for me i think i'd like to touch back on to um what other situations are like the elizabethan play people like the elizabethan cast um the theater company that's what it is sorry uh so what other situations are like the elizabethan theater company in that the culture is so strong and so vital to participation that um you can just wing it like you can know a lot of things just by queuing off of others so i had mentioned jazz i don't know what else um what other i'd like to look at that a little bit more because it's that that was the example that they gave right this this diachronic cognition so it would be cool to look at that a little more deeply yes especially in like the kind of online context um yes um agreed about that yeah you brought up music and jazz and about culture where does what is culture of different kinds and where does that fit into this ability of people to have joint action in what ways or in what senses is culture about shared ways of thinking orthodoxy heterodoxy in what ways is culture about joint action orthopraxis and heteropraxis um so i was thinking also about action um and like action offloading uh uh because just like in terms

of a scaled system like sure if cells do their own actions like they eat food and they you know secrete nutrients and you know they like communicate with each other and whatever but but in terms of like a nested system um like the cells don't have to go find water the water just comes to them right
like like whereas a human it's like one of the most vital things that we could ever do is find
water like clean drinking water super necessary for life so we have to do that i mean we've
offloaded that into our culture with running water but not every human um but i just think
about we're acting on behalf of organ systems cell systems organelles etc so i think about action
offloading so what does that look like um yes and how how akin to cognitive offloading or uploading is
that yes in what ways is the cognitive offloading and uploading that of perception in what ways is it
offloading action like if you write a script that then does some daily task that could be seen as
like offloading a repetitive action and that's something that we're only unlocking with computers
but it's kind of like a and it's even referenced in the idea of a of a daemon in computers it's like the
magicians familiar it's this little trainable or learning entity that is able to carry things out
according to the will of the magician in a semi-autonomous way it's not just computers
though we've automated some tasks for us through the years through the ways of like even just with
running water like we you know that's an automated task that used to be a big portion of
your daily life as a human like going to find water um and i mean washing machines dishwasher we've all
the all the things i mean we've been automating life for forever so now it's just becoming more
sophisticated now perhaps we're automating our cognitive life like with gpt3 i think that that's
like a a different kind of yes we're it's um approaching automation at the semantic and linguistic
level whereas previously it could be like at the task the physical task performance level could be
off-sourced or uploaded in a sense this is like rapidly rising in its cognitive complexity um
like a typewriter is a mere active entity it's part of extended cognition type type type type type
but unless it gets jammed one is not going to be too surprised by how it performs um and that's like
the famous turing quote from back in the day and someone said like um how can it have intelligence or um
if it can't give a surprising answer and uh and then turing said something like computers surprise me every day
so is that going to be your measure how surprised you are um let's just want to look at what do you
want to look at though in the next one look at um i want to look at the partition question that dave
raised and also the earlier question from dave when he wrote about the result of cognitive offloading it
was um it was more about cognitive uploading he clarified in the chat but i know i was thinking
yeah i was thinking about the partition too like when i was talking about i meant to say partition
in there because it's a temporal partition going from this generation to the next or is it like a
cognitive or like a organismal partition between like me and you or me and my daughter or this group
in that group um the partition is the fuzziness of the partition always makes it interesting okay what
else let's let's that's it let's read his question so in some papers other than this one
so here's the the niche of papers citing each other we're in an adjacent regime of an attention
but it's it's relevant the point comes up and this comes up indeed in discussion anatomical or

mechanical blankets so at the cellular level this would be like the lipid bilayer cell membrane at the organismal level this would be like the epithelial so anatomical gross or micro mechanical blankets sometimes coincide with functional psychological blankets and sometimes not so two brain regions that are separated but have an interface it might be the case that their function could be ascribed with a certain partition that reflected the partitioning topologically of the anatomical components or it could totally not be that way so sometimes anatomical mechanical blankets coincide with functional psychological blankets and sometimes not insert all the asterisks that we've ever had about realism and instrumentalism but just we're going to plunge ahead what is the relation among that distinction between the two kinds of blanketing this one more of a um embodied enacted realist flavor and this one more of a functionalist
flavor
so what is the relation among a b and c here so what is it's a relation of relation of relations as per usual we're relating a b and c where a is the relationship between these kinds of blankets b
is the hierarchical and heterarchical nesting of blankets and c is the relations between or among niches and the particles that upload functionality to those niches
and the other ones that are not being able to do so what do you think blue about when they say that the answer is in the question
like the way that the question is asked contains the seeds of an answer what do you think about that um i mean are you talking about like how i would you know cognitive offload information as an undergraduate just because i could tell by the way that the question was asked yeah like i think i did that um like i think i looked for i looked for the answer in the question quite often as long as it's a multiple choice question that was pretty simple or if it wasn't there was often somewhere else in the test in a different question so like okay so yeah i did that um but i think here
there's definitely some
um answering here happening like that distinction what is that like i hate those i hate those words i hate i hate those words as i use like a the similar like referential what is the name of like the part the part of speech that it is this that those these you know those those words words that are um
um it's a pronoun oh it's an article it's an article because the pronoun is like he she it so it's the articles so the articles really bother me i think that um that distinction the distinction between anatomical mechanical blankets or the functional psychological blankets yes i'll write it out i believe it's the distinction between anatomical mechanical and functional psychological
those so it should be a plural article
that's what was throwing me um yeah anyway the nesting of blankets is always something like i am down to talk about but but here i don't know does the does the distinction need to be drawn
if you're talking about nesting of blankets how can you
make distinctions between different types of blankets i don't know maybe that's just me
what do you think we explored some aspects of it when you mentioned the multi-scale
approach where like we can think of nesting as the the bigger nested system whether it hosts only one

active entity within it or multiple laterally interacting entities
the bigger nesting so putting a aside for now the bigger nested system can be seen of as the niche
or the subunit or subunits inside of it so for sure this b and c can be connected because the nested system
it's niche it's environment what its actions are going into and what it is getting sense information from
is the niche and that is like the larger nested system now how do nested systems
using that as a shorthand for both b and c connect to this
structural and functional perspective
on systems are nested systems when we talk about nested systems in what ways can be yes and is it a
structural nesting like the word is inside the sentence inside of the paragraph or in what ways is it a
functional nesting where the functional nesting of concepts doesn't necessarily have to follow
from the nesting of like sentences on the page i think that's a very challenging but important question
and then also when you mentioned like the undergrad example um and we've been talking a lot about test
taking right here it made me think about how in undergrad or potentially even earlier yes it was
possible to like kind of reverse engineer and so like the answer is in the seeds of the question
is kind of like a game to play where you're just trying to maximize your score on some test but then
you get rugged in grad school where it is still true that the question that you're asking to nature to
the system contains the seeds of the answer like you're only going to get the data for your experiment
of the kind that you measure on the number that you actually measure it from with the conditions that
you set them up to be in but if one tries to p hack nature or if one tries to do that sort of
undergrad i'm going to maximize i mean these two groups must be different right so i'm going to try to
measure them with a thumb on the scale so that they come out to be different that becomes
tantamount to basically non-rigorous non-reproducible science so it just shows i think also to dean's
continued themes of like in the lab in the classroom and then out so the highest test score
is not necessarily the highest performer on that out in the wild that's towards the application side
nor necessarily the best teacher as we've been discussing and so it's sort of like there's like
learning and development and sometimes it feeds back on itself with teaching and other times it just
plunges ahead with application to all these different phenomena of learning teaching and applying
to the same to the same to the same to the same to the same to the same to the same to the same
are they featuring or highlighting their regime of attention on different parts of the partitioning
so that's like where i can't ever close my instrumentalist eye um it is in the partition uh
point because the partition is just wherever you put it
is in the real world um um um um um um um um um um um um um um um um um um um um um um um um um um
make claims about realism i see right i mean but even if you want like when i investigate realist
claims like if i look for a partition like what partitions me from the rest of the world it's like
my skin right so like my skin is the the boundary it's the markov blanket that holds me in and the
world out but like i'm losing pieces of my skin every day it gets cut open and so so then am i
is my partition broken or like you know am i somehow not the same person i don't know so for

me finding this like boundary that's like solid and never changes is not doesn't sit right so that's

why like i always have to remind myself it's an instrument so i can never ever lose my instrumentalist

perspective and get into full realism when i think about the blanket so um it's it's non non-separability

it's you know this is goes back to the quantum chris fields you know we're non-separable

fun time so i think in uh dot two let's just write a few notes down and then close out so i think in

dot two we can look over some formalisms and see how the equation is connected

we can think about what we discussed about the fastest time scales perception action learning

teaching applying etc how do these different modes of engagement with extended active inference systems

different states

How do we go from negotiating, trusting, discomforting,

communicating, to the right graphical structure of a generative model

that we can use in our standard ACTEMP routines,

as patiently reminded we all are by Carl on those points?

Very cool.

Okay.

Fun times.

Thanks a lot, Dean, for joining.

Yeah, Dean, hope you feel better.

Yep, absolutely.

And Blue, thanks a ton.

It was a fun discussion.

Thanks, everybody who's in the live chat, Dave and Tucker and Steven.

And everybody is welcome to join us next week for 41.2 on 13th and in the papers beyond.

So thank you all.

See you later.

Peace.

Bye.